BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 285 · Network security, proxies and firewalls

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Can network packets travelling between the same two endpoints take different routes?

Q2. What is the transport-layer role of TCP?

Q3. What does a rule ‘allow if staff AND authenticated’ mean if authenticated is false?

Q4. Why can an open port be useful but also increase risk?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

firewall — Security control that permits or blocks network traffic according to rules.

forward proxy — Intermediary that makes requests on a client's behalf.

filter rule — A condition and action defining whether traffic is permitted.

cache — Stored copy of suitable data for reuse.

defence in depth — Use multiple complementary security controls rather than trusting one.

New learning

Check a firewall rule — A firewall applies configured rules to traffic according to properties such as direction, source/destination, ports and state. A basic rule example can allow TCP port 443 but deny 22.

Different request, different decision — A TCP connection to port 22 is denied by the illustrated policy. A firewall rule is not automatically a guarantee that the permitted service is safe.

A proxy is an intermediary — A forward proxy makes requests on behalf of clients, potentially enforcing policy or caching suitable resources. HTTPS content remains encrypted unless a controlled interception mechanism is in place.

Worked example

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permitted.

1. 443 — An inbound TCP connection destined for port 443 matches the allow rule and is permitted.

2. 22 — An inbound TCP connection destined for port 22 matches the deny rule and is blocked.

3. Scope — An allow rule permits traffic; it does not prove that an application or connection is harmless.

We Do · Practise with guidance

A school uses a forward proxy to restrict browsing. Explain the route for a permitted request and for a blocked site.

Allowed resource

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.

Blocked resource

If the proxy's policy denies the destination, it does not forward the request to that site; the student receives a blocking response.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A firewall allows inbound TCP 443 but denies inbound TCP 22. Is an inbound connection to destination port 22 allowed? Explain. [2]

Question 2

Name and explain two different functions of a forward proxy in a school network. [4]

Question 3

Why is the claim ‘a firewall makes all allowed traffic safe’ incorrect? Explain two risks that can remain on allowed connections and identify two additional safeguards. [6]

Exit Ticket · Answer before checking

A forward proxy and firewall are identical because both can restrict access.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Yes, routing decisions and changing network conditions can lead to different paths.

Q2. A reliable, ordered byte stream between endpoints.

Q3. Deny; AND requires both conditions true.

Q4. It permits needed service connections, but exposes that service to network-originated attacks or probing.

You Do pages · Independent questions

Q1. [2] No, the connection is blocked [1], because inbound TCP destination port 22 matches the deny rule [1].

Q2. [4] Accept any two appropriately explained functions. Examples: apply browsing rules [1] to deny disallowed destinations before contacting them [1]; cache suitable responses [1] to reduce repeat external downloads [1]; or log request metadata [1] to support authorised monitoring or troubleshooting [1]. Maximum two functions × 2 marks.

Q3. [6] Allowed applications can contain exploitable vulnerabilities [1], with a relevant explanation such as an unpatched service being attacked on an allowed port [1]. Permitted traffic can also contain phishing or malicious content [1], which may still reach a browser or user [1]. Identify two additional safeguards with purpose, e.g. patching to close known flaws [1] and strong authentication or monitoring to reduce misuse or detect incidents [1]. Credit other valid distinct risks and defences.

Exit Ticket · Model answer

False. A proxy acts as an intermediary for requests; a firewall enforces rules on traffic.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: